MTH209: Assignment 5

Transformation of Gaussian

Your task is to write a function that takes a vector X that is a realization from $N(\mu, \Sigma)$ and returns a realization from $N(\mu, \Sigma^{-1})$ where $\mu \in \mathbb{R}^d$ and Σ is a $d \times d$ positive-definite matrix.

```r
# x: a vector of length d lying in the d-dim space
# mu: mean vector in R^d
# Sigma: d x d covariance matrix

gauss_convert <- function(x, mu, Sigma)
{
  d <- length(x)
  ...
  ...
  transform <- numeric(length = d)
  ...
  return(transform) # output must be a vector of length d
}
```

I will take your function and test the function for my own choice of the input x, μ, Σ.

INSTRUCTIONS: Upload the file with the name `assignment5.R` on helloIITK. The file must contain ONLY the function and nothing else